Question	Answer	Marks	Partial Marks
4(a)(i)	$\frac{1}{2}\times\frac{4}{3}\times\pi\times5.6^3$	**M1**	
	367.8... to 367.9	**A1**	
4(a)(ii)	3.06 or 3.060 to 3.061...	**4**	**M1** for 0.8 × 368 [= 294.4] **M2** for [r^2 =] $\frac{their\ 294.4}{10\pi}$ oe or **M1** for $\pi r^2 \times 10 = their$ 294.4 oe
4(b)(i)	44[.0] or 43.98 to 43.99 nfww	**5**	**B2** for [slant height =] $\frac{25}{4}$ oe or **M1** for [l^2 =] $6^2 + 1.75^2$ oe **M2** for $\pi\times1.75\times their\ l+\pi\times1.75^2$ or **M1** for $\pi\times1.75\times their\ l$ or $\pi\times1.75^2$
4(b)(ii)(a)	SF = $\frac{1}{4}$ oe soi	**B1**	
	$\frac{1}{3}\pi\times1.75^2\times6-\frac{1}{3}\pi\times their\ 0.4375^2\times1.5$ **OR** $\frac{1}{3}\pi\times1.75^2\times6\times\left(1-\left(\frac{1}{4}\right)^3\right)$ oe	**M2**	**M1** for $\frac{1}{3}\pi\times1.75^2\times6$ or $\frac{1}{3}\pi\times their\ 0.4375^2\times1.5$ **OR** **M1** for $1-\left(\frac{1}{4}\right)^3$ oe
	18.94 or 18.939 to18.944…	**A1**	
4(b)(ii)(b)	95 final answer	**3**	**B2** for 94.5 or 94.69 to 94.722 OR **M2** for $18.9\times10^3\div200$ oe or **M1** for 18.9×10^3 or $200\div10^3$ or figs 189..÷ 200 or 18.9.. ÷ figs 2
5(a)(i)	–3	**1**	
5(a)(ii)	6.2 to 6.4 oe	**2**	**M1** for 3 seen or used
5(b)	$y = 5 - 3x$ ruled	**2**	**B1** for $y = 5 - 3x$ soi or ruled line with gradient – 3 or with y – intercept at 5 (but not $y = 5$) or **B1FT** for incorrect line equation/expression shown in working and *their* line correctly drawn
	– 0.3 to – 0.2 1.65 to 1.8	**2**	**B1** for each, dep on $y = 5 - 3x$ drawn or **FT** *their* line provided equation/expression shown in working, dep on B1FT for line

Question	Answer	Marks	Partial Marks
5(c)	Tangent ruled at $x = -2$	**1**	**B1** for correct tangent
	–4.5 to –2.5	**2**	**Dep** on B1 for tangent or close attempt at tangent at $x = -2$ **M1** for rise/run **also dep on** tangent drawn or close attempt at correct tangent Must see correct or implied calculation from a drawn tangent
5(d)(i)	8, 4, 0.25 oe	**3**	**B1** for each
5(d)(ii)	Correct graph	**3**	**B2FT** for 6 or 7 correct plots or **B1FT** for 4 or 5 correct plots
5(d)(iii)	1.8 to 1.9	**1**	
6(a)	40.5 or 40.45[8..] or 40.46 nfww	**4**	**M1** for 25, 32.5, 37.5, 50, 80 soi **M1** for Σft **M1 dep** for their $\Sigma ft \div 120$
6(b)	Fully correct histogram	**4**	**B1** for each correct bar If 0 scored, **SC1** for frequency densities of 5.4, 4.2, 0.8 and 0.45 seen
7(a)	[y =] $4x + 5$	**3**	**B2** for answer [y =] $4x + c$ oe (c can be numeric or algebraic) OR **M2** for $\frac{y-9}{x-1} = \frac{9-(-3)}{1-(-2)}$ oe OR **M1** for $\frac{9--3}{1--2}$ oe or for **M1** for correct substitution of (–2, –3) or (1, 9) into $y = (their\ m)x + c$ oe
7(b)	76[.0] or 75.96...	**2**	**M1** for tan[ ] = 4 oe